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ON SIMULATION FROM INFINITELY DIVISIBLE DISTRIBUTIONS

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Abstract

A general method based on a limit theorem for generation of random numbers from infinitely divisible distributions with essentially given Lévy measure is studied. Some classes of infinitely divisible distributions that appear in a natural way in this context are paid particular attention.

SHOT-NOISE DISTRIBUTION; SELF-DECOMPOSABILITY; GENERALIZED GAMMA CONVOLUTION; STABLE DISTRIBUTION; NORMAL APPROXIMATION

1. Introduction

There is a huge literature on how to generate random numbers from different distributions; many references can be found in, e.g., the recent book by Kennedy and Gentle (1980). However, it seems as though general methods for simulation from infinitely divisible (ID) distributions given by their transforms are not available in the literature. A main purpose of the present paper is to fill that gap. A method is described that works for any ID distribution with more or less explicitly given Lévy measure. The method is not based on inversion of transforms. Instead, it utilizes an old but perhaps not very well-known fact that essentially any ID distribution without normal component can be represented as the marginal distribution for a (generalized) shot-noise process with a suitable response function. Shot-noise processes appear in many fields, e.g., in insurance mathematics if inflation is not neglected.

This is mainly a theoretical paper and some particularly interesting classes of ID distributions are given special attention. The purpose is to point out and indicate possibilities but not to investigate in detail how the simulation method, in its various forms, works in practice. Certainly its efficiency varies considerably. To avoid too many technical details essentially only ID distributions on $[0, \infty)$ are considered. The reader is assumed to know basic facts about ID. A section on ID and shot-noise processes can be found in Moran (1968), pp. 419–433. See also Feller (1971), Chapter 6.

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2. Some fundamental facts

The Laplace (–Stieltjes) transform (LT) of an ID distribution on $[0, \infty)$ is given by

$$\varphi(s) = \exp\left\{-as + \int_{(0,\infty)} (e^{-sy} - 1)N(dy)\right\},$$

where $a \geqq 0$ and the non-negative Lévy measure $N(dy)$ satisfies $N((1,\infty)) < \infty$ and $\int_{(0,1]} yN(dy) < \infty$. Here a is the left-extremity of the distribution and is only a translation parameter. In this paper we always assume that $a = 0$.

If $\lambda = N((0,\infty)) < \infty$, we may introduce a d.f. G on $(0,\infty)$ by $G(x) = \lambda^{-1}N((0,x])$. Then $\varphi(s) = \exp\{\lambda(\gamma(s) - 1)\}$, where γ is the LT of $G(dy)$. A random variable with LT φ is given by $X = \sum_{i=1}^{\nu} Y_i$, where the Y_i's are i.i.d. with d.f. G and ν is an independent Poisson (λ) random variable. This gives us an obvious method to simulate from the given ID d.f. (F, say) if λ is not too large and if the Y_i's can be simulated easily.

If $\lambda = \infty$, we may approximate N by N_ε defined by $N_\varepsilon(A) = N((\varepsilon,\infty) \cap A)$ and simulate values X_ε from the ID d.f. F_ε corresponding to N_ε. If ε can be chosen so that $\lambda_\varepsilon = N_\varepsilon((0,\infty))$ is not large and so that the 'error variable', e_ε, with LT $\exp\{\int_{(0,\varepsilon]} (e^{-sy} - 1)N(dy)\}$ and independent of X_ε, is relatively small (in probability), then $X = X_\varepsilon$ is an approximate random number from F that can be simulated easily. Often there is also a possibility of using as substitute for e_ε a random number from a normal distribution with mean $\int_{(0,\varepsilon]} yN(dy)$ and variance $\int_{(0,\varepsilon]} y^2 N(dy)$.

The above method is not very flexible since ε must be fixed in advance and it may be difficult to simulate from G. The method that is now presented is much more general and flexible.

Let $Z(u)$, $u > 0$, be a family of non-negative independent random variables. Let T_i, $i = 1, 2, \ldots$, be the points (in increasing order) in an independent Poisson point process of rate λ (Po (λ)-process) on $(0,\infty)$. We set

$$X = \sum_{i=1}^{\infty} Z(T_i). \tag{2.1}$$

Here X equals $X(0)$ for a (generalized) shot-noise process $X(t) = \sum_{T_i' \leqq t} Z(t - T_i')$, where $\{T_i'\}_{-\infty}^{\infty}$ are the points in a Po(λ)-process on $(-\infty,\infty)$. Therefore we call the distribution of X a *shot-noise distribution.*

Set $X_T = \sum_{T_i \leqq T} Z(T_i)$, where T is a truncation point. Its LT is given by

$$E[\exp\{-sX_T\}] = \exp\left\{\lambda \int_{(0,T)} (\zeta(s;u) - 1)\, du\right\},$$

where $\zeta(s;u) = E[\exp\{-sZ(u)\}]$; this can be proved in at least two ways. One proof is based on a conditioning of the number of points in $(0, T]$, and the

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other, more unstringent one, is based on an approximation (in distribution) of X_T as $\sum_{k=1}^{[Tn]} Z(k/n)I_k^n$, where $I_k^n = 1(0)$ if there are (no) points in $(k-1/n, k/n]$. It follows that X has LT

$$\varphi(s) = \exp\left\{\lambda \int_{(0,\infty)} (\zeta(s;u)-1)\,du\right\}. \tag{2.2}$$

We assume that $X<\infty$ a.s. (otherwise $X=\infty$ a.s.; this follows from Kolmogorov's zero–one law and the fact that the process X_T, $T>0$, has independent increments). It is obvious that X is ID. (The same result holds for an inhomogeneous Poisson process but we need not use that.) Let $Z(u)$ have d.f. $H(y;u)$. Changing the order of integration, we may rewrite (2.2) as

$$\varphi(s) = \exp\left\{\lambda \int_{[0,\infty)} (e^{-sy}-1)\left(\int_{(0,\infty)} H(dy;u)\,du\right)\right\}.$$

Of course, the measure $H(dy;u)$ must as a function of u satisfy certain regularity conditions.

Now we consider simulation from an ID d.f. F with Lévy measure $N(dy)$. Suppose we can find a simple family of d.f.'s $H(y;u)$ on $[0,\infty)$ and a λ such that on $(0,\infty)$

$$\lambda \int_{(0,\infty)} H(dy;u)\,du = N(dy) \tag{2.3}$$

or equivalently, for $x>0$,

$$\lambda \int_{(0,\infty)} \bar{H}(x;u)\,du = \int_{(x,\infty)} N(dy) = \bar{N}(x), \tag{2.4}$$

where $\bar{H} = 1-H$. Then we simulate points T_i in a Po(λ)-process by, e.g., adding independent exponential random numbers and after that values $Z(T_i)$ from the d.f.'s $H(x;T_i)$ and set $X=\sum_{i=1}^{\infty} Z(T_i)$. Then X has the desired distribution. If the sum converges rapidly, only a few terms are needed to get a good approximative value for X. In Section 4 we discuss how this method should be used in practice. In the next section, however, various classes of shot-noise distributions are studied.

3. Classes of shot-noise distributions

3.1. For a given ID distribution with Lévy measure N one can always find $H(y;u)$ so that (2.3) is satisfied as we shall see in this section. However, $H(y;u)$ is not unique.

The d.f.'s of the form $H(y;u) = H(y/g(u))$, where H is a d.f. on $[0,\infty)$ and g is a non-negative function, is a family of particular interest. In this case

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$Z(u) = g(u)V(u)$, where the components of $V(u)$ are i.i.d. with d.f. H. It is no loss of generality to assume that $V(u) > 0$ (why?). The function g is called the response function and H the shot-distribution. Rearranging g, we may assume it to be non-increasing.

The relation (2.4) can now be written

$$(3.1) \quad \lambda \int_{(0,\infty)} \bar{H}(x/g(u))\, du = \lambda \int_{[0,\infty)} \bar{H}(x/v) g^{-1}(dv) = \bar{N}(x), \qquad x > 0,$$

where $g^{-1}(dv)$ is the measure that the Lebesgue measure μ through g induces on $[0, \infty)$. If $V(u) \equiv 1$, i.e., if H is degenerate at 1, then (3.1) reduces to

$$(3.2) \qquad \mu\{u; g(u) > x\} = \bar{N}(x)/\lambda, \qquad x > 0.$$

Since g is non-increasing, the left-hand side of (3.2) is an 'inverse' of g:

$$g^{-1}(x) = \sup\{u; g(u) > x\}.$$

Of course, for any N and λ a non-increasing g can be found so that (3.2) is satisfied. It suffices to set

$$g(u) = \mu\{x; \bar{N}(x)/\lambda > u\} = \sup\{x; \bar{N}(x)/\lambda > u\}$$

with the convention that the supremum is 0 if the set is empty. We see that every ID d.f. on $[0, \infty)$ (with left-extremity 0) is a shot-noise distribution with a degenerate shot-distribution H and non-increasing response function g.

It follows from what is presented below that for given N and H (or g) it is not always possible to find g (or H) so that (3.1) is satisfied.

3.2. Let us consider the class of shot-noise distributions that is generated by exponential shot-distributions H. Since g can absorb a possible scale parameter, we may set $\bar{H}(x) = e^{-x}$. Then (3.1) becomes

$$\lambda \int_{[0,\infty)} e^{-x/v} g^{-1}(dv) = \bar{N}(x),$$

and we see that $\bar{N}(x)$ is completely monotone, i.e., its derivatives have alternating signs. Equivalently, N is absolutely continuous with a completely monotone density $\boldsymbol{n}$. Conversely, by using Bernstein's theorem that every completely monotone function is the LT of a non-negative measure, we easily see that every ID d.f. F with completely monotone Lévy density $\boldsymbol{n}$ (and left-extremity 0) is a shot-noise distribution for an exponential shot-distribution H and appropriate (non-increasing) response function. As noticed in, e.g., Kent (1982) the class of ID distributions with completely monotone Lévy density coincides with the class $\mathcal{T}_2$ of *generalized convolutions of mixtures of exponential distributions* studied in Bondesson (1981). This class can be described as the class of distributions that are obtained by taking weak limits of

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finite convolutions of mixtures of exponential distributions. A distribution in $\mathcal{T}_2$ is most conveniently described by a non-negative measure Q on $(0, \infty)$. We have that $\boldsymbol{n}(x) = \int_{(0,\infty)} e^{-\tau x} Q(d\tau)$ and g should be an inverse of the function $x \to \lambda^{-1} \int_{(0,\, 1/x]} \tau Q(d\tau)$. By using the fact that a distribution in $\mathcal{T}_2$ is a mixture of exponential distributions iff $Q((a, b]) \leqq b - a$ for every interval $(a, b]$, one can identify the mixtures of exponential distributions as the shot-noise distributions that are generated by response functions g with $e^{\lambda u} g(u)$ non-increasing.

The class $\mathcal{T}_2$ can also be described as the class of distributions of weak limits of finite convolutions of distributions of random variables of the form $\sum_{i=1}^{\nu} Y_i$, where the Y_i's are i.i.d. exponential random variables and ν is an independent Poisson random variable. This description explains essentially why $\mathcal{T}_2$ appears in the context here. If H is a mixture of exponential d.f.'s, the shot-noise distribution is still in $\mathcal{T}_2$. However, for H a gamma d.f. with shape p, the possible shot-noise distributions are characterized by having a Lévy density $\boldsymbol{n}$ such that $x^{1-p}\boldsymbol{n}(x)$ is completely monotone. The corresponding class coincides (modulo a translation) with the class $\mathcal{T}_{p+1}$ in Bondesson (1981); cf. Berg and Forst (1982).

3.3. Let us now look at the class of shot-noise distributions that is generated by an exponential response function $g(u) = Ce^{-\rho u}$, $\rho > 0$. This is the most natural response function (that gives a Markovian shot-noise process). It is no restriction to set $C = \rho = 1$ since $\{\rho T_i\}_1^\infty$ are points in a $\mathrm{Po}(\lambda/\rho)$-process. After some calculation (3.1) can then be written as

$$\lambda \int_{(x,\, \infty)} y^{-1} \bar{H}(y)\, dy = \bar{N}(x). \tag{3.3}$$

Equivalently, N is absolutely continuous and

$$\lambda \bar{H}(x) = x\boldsymbol{n}(x). \tag{3.4}$$

It follows easily that a necessary and sufficient condition for the existence of a λ and an H such that (3.4) be satisfied is that N be absolutely continuous and $x\boldsymbol{n}(x)$ non-increasing with a finite limit as $x \downarrow 0$. The condition that $x\boldsymbol{n}(x)$ be non-increasing is exactly the condition that the d.f. F be *self-decomposable*; see, e.g. Feller (1971), pp. 588–590. So every shot-noise distribution with exponential response function is self-decomposable and, as is not hard to see, every self-decomposable distribution with left-extremity $\geqq 0$ is a limit of shot-noise distributions with exponential response function. This solves a problem mentioned by Lawrance (1982).

Setting $g(u) = u^{-\beta}$, $\beta > 1$, we easily see from (3.1) that $\bar{N}(x)$ is proportional to $x^{-1/\beta}$ for any H. Thus, in this case, the shot-noise distribution is stable for any possible shot-distribution.

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3.4. Let us now set $\bar{H}(x)=e^{-\tau x}$ and $g(u)=e^{-u}$. From (3.4) we get $\boldsymbol{n}(x)=\lambda x^{-1}e^{-\tau x}$. Thus the shot-noise distribution is a gamma distribution with shape λ and scale $1/\tau$ since

$$(1+as)^{-\lambda}=\exp\left\{\lambda\int_{(0,\infty)}(e^{-sy}-1)\frac{1}{y}e^{-y/a}\,dy\right\}.$$

This result, old and difficult to attribute to a specific person, has appeared in stochastic hydrology and was discussed by Lawrance (1982) in connection with simulation for AR-1 processes. It is remarkable because, in spite of the popularity of the gamma distribution, there seems to be a lack of models of real phenomena with variables that are gamma distributed with shape parameters that can take any positive value. It is also remarkable that the shot-noise distribution is a convolution power of the (exponential) shot-distribution. One can easily show that for $g(u)=e^{-u}$ this is true precisely for shot-distributions that have LT of the form $(1+as^{\alpha})^{-1}$, $0<\alpha\leqq 1$, $a\geqq 0$ (mixtures of stable distributions with exponential mixing distribution).

Remark. The above shot-noise representation of the gamma distribution also leads to an improved representation of Ferguson's (1973) Dirichlet processes. If we let G be a d.f. and set

$$X(t)=\sum_{i=1}^{\infty}\exp\{-T_i\}V_iI_{(0,G(t)]}(U_i),$$

where $\{T_i\}_1^{\infty}$ are points in a Po (λ)-process on $(0,\infty)$, the V_i's i.i.d. exponential random variables (mean 1), the U_i's i.i.d. $U(0,1)$-variables, and I_A the indicator function of a set A, then the process $X(t)$ has independent increments $X(t_2)-X(t_1)$ that are gamma distributed with shape $\lambda(G(t_2)-G(t_1))$.

3.5. For $g(u)=e^{-u}$ a more general class of shot-noise distributions than the gamma ones is obtained by letting the shot-distribution be a mixture of exponential distributions, i.e.,

$$\bar{H}(x)=\int_{(0,\infty)}e^{-\tau x}M(d\tau), \tag{3.5}$$

where M is the mixing probability measure. (We assume that $\int_{(0,1)}|\log\tau|\,M(d\tau)<\infty$; the shot-noise distribution is proper iff this condition holds.) By (3.4) we have

$$x\boldsymbol{n}(x)=\lambda\int_{(0,\infty)}e^{-\tau x}M(d\tau).$$

Together with Bernstein's representation formula for completely monotone functions, this result easily shows that an ID distribution (with left-extremity 0)

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is a shot-noise distribution for an exponential response function and a shot-distribution of the form (3.5) iff $x\boldsymbol{n}(x)$ is completely monotone and has a finite limit as $x \downarrow 0$. The LT of such a shot-noise distribution can be written as

$$\varphi(s) = \exp\left\{\int_{(0,\infty)} \log\left(\frac{1}{1+(s/\tau)}\right) U(d\tau)\right\},$$

where here $U(d\tau) = \lambda M(d\tau)$. We see that the class of these shot-noise distributions coincides with Thorin's (1977) class of *generalized gamma convolutions* (g.g.c.) with left-extremity 0 and finite (Thorin) measure U. It follows easily that every g.g.c. is a weak limit of g.g.c.'s of the special type. Of course, it is probabilistically fairly obvious that mixing on the shot-distribution side should correspond to convolution on the shot-noise distribution side.

Obviously every g.g.c. is self-decomposable. Since $\boldsymbol{n}(x)$ is completely monotone whenever $x\boldsymbol{n}(x)$ is, every g.g.c. belongs to $\mathcal{T}_2$. A self-decomposable distribution in $\mathcal{T}_2$ need not be a g.g.c.

3.6. Let us now briefly consider shot-noise distributions on $Z_+ = \{0, 1, 2, \ldots\}$. Clearly in this case the shot-distribution must also be a distribution on Z_+. Let $Z(u)$ have generating function $\zeta(s; u) = E[s^{Z(u)}]$, $0 \leqq s \leqq 1$. Then, in complete analogy with (2.2), the shot-noise variable $\sum_{i=1}^{\infty} Z(T_i)$ has generating function

$$\varphi(s) = \exp\left\{\lambda \int_{(0,\infty)} (\zeta(s; u) - 1)\, du\right\}. \tag{3.6}$$

We now show that any negative binomial distribution can be obtained as a shot-noise distribution in an easy way.

Let $Z(u)$ be geometrically distributed with parameter $p(u) = p/(p + qe^{-u})$, where $p \in [0, 1]$ and $q = 1 - p$. We may call $p(u)$ a response parameter. The corresponding generating function is $\zeta(s; u) = p(u)/(1 - q(u)s)$, where $q(u) = 1 - p(u)$. Then (3.6) and some calculation show that the generating function of the shot-noise distribution is $(p/(1 - qs))^{\lambda}$. This is the desired result.

If p is stochastic, the distribution of $Z(u)$ is instead a mixture of geometric distributions and the shot-noise distribution is a *generalized negative binomial convolution* with left-extremity 0 and finite spectral measure V; cf. Bondesson (1979).

3.7. Let us finally in this section consider ID distributions on R without going into details. Let $\{T_i\}_{-\infty}^{\infty}$ be points in a Po(λ)-process on R. Set

$$X_I = \sum_{T_i \in I} Z(T_i), \tag{3.7}$$

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where $Z(u)$, $u \in R$, is a family of independent random variables and $I=[-a, b]$ an interval. Setting $Z(u)=g(u)$, where $g(u) \geqq 0 (\leqq 0)$ and $g \downarrow (\uparrow)$ for $u>0$ $(u<0)$, choosing g appropriately, and letting a and b tend to ∞ in a suitable way, we can obtain (up to a translation) any ID distribution without normal component as limit distribution. (Kolmogorov's zero–one law implies that moreover X_I almost surely converges. However, it is necessary to truncate suitably since $\sum_{T_i>0} g(T_i)$ may be divergent. If the Lévy measure is symmetric we may take $a=b$. We may also set $a=b$ in certain asymmetric cases if we instead let the Poisson process have different rates on the two sides of the origin. For example, stable distributions with index α, $1<\alpha<2$, can be obtained in this way if we set $g(u)=c_+u^{-1/\alpha}$ for $u>0$ and $g(u)=-c_-|u|^{-1/\alpha}$ for $u<0$ with c_+, c_- and the rates suitably chosen.

If $Z(u)=e^{-|u|}V(u)$, where the family of random variables $V(u)$, $u \in R$, are i.i.d. and without too long tails of their common distribution, (3.7) converges absolutely almost surely to a random variable X. Since $e^{-|u|}$ is a symmetric function we may write X as

$$X=\sum \exp\{-T_i\}V_i$$

where $\{T_i\}$ are now points in a Po(2λ)-process on $(0, \infty)$ and $V_i=V(T_i)$. Then we may write X as X^+-X^-, where $X^+=\sum \exp\{-T_i'\}V_i'$ and $X^-=\sum \exp\{-T_i''\}V_i''$ and $\{(T_i', V_i')\}=\{(T_i, V_i); V_i>0\}, \{(T_i'', V_i'')\}=\{(T_i, -V_i); V_i<0\}$. Here $\{T_i'\}$ and $\{T_i'\}$ are points in independent Poisson processes of rates $2\lambda P(V_1>0)$ and $2\lambda P(V_1<0)$, respectively. Since we have already found a description of the possible distributions of X^+ and X^- in terms of self-decomposability, we have also a description of the class of possible distributions of X. The class is a proper subclass of the class of self-decomposable distributions but its weak closure equals the last class.

A particularly interesting case is when V_i' and V_i'' have distributions which are mixtures of exponential distributions. Then X is a difference of two variables whose distributions are g.g.c.'s with finite Thorin measure and left-extremity 0. Taking also weak limits we get precisely the class of *extended generalized gamma convolutions* introduced by Thorin (1978).

4. Simulation

4.1. Suppose we want to simulate from an ID d.f. F on $[0, \infty)$ with left-extremity 0 and Lévy measure N. We know that for some family of d.f.'s $H(y; u)$, $u>0$, on $[0, \infty)$,

$$X=\sum Z(T_i) \tag{4.1}$$

is a random number from F if T_i, $i=1, 2, \ldots$, are simulated points from a Po(λ)-process on $(0, \infty)$ and $Z(T_i)$ is a simulated value from $H(y; T_i)$. In particular we may set $\lambda=1$ and $Z(u)=g(u)=\sup\{x; \bar{N}(x)>u\}$.

Of course, there is no loss in setting $\lambda=1$ but the d.f.'s $H(y; u)$ can be

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chosen in more or less good ways. Their choice should be dictated by two demands which may often be in conflict:

A. It should be easy to simulate the values $Z(T_i)$;

B. The sum giving X should either converge rapidly or so slowly that the tail of the sum can be replaced by a normal random number.

We shall consider examples later on. An interesting theoretical problem is whether appropriately chosen non-degenerate random variables $Z(u), u>0$, can give better speed of convergence than the corresponding degenerate random variables $Z(u)(=g(u)$ above). The answer would have been affirmative if for a distribution with infinite Lévy measure, $Z(u)$ could have been chosen so that $Z(u)\equiv 0$ for all u sufficiently large. However, it follows easily from (3.1) that that is imposssible.

Assuming that the choice of $H(y;u), u>0$, has been made, we face the problem how to calculate or get a good approximate value of the sum (4.1). There are several ways of truncating the sum:

1. Sum until several consecutive partial sums are so close that the tail of the sum can be judged to be negligible;

2a. Sum over all $T_i \leqq T$, where T is a number chosen so that the mean μ_T and variance σ_T^2 of the tail $e_T=\sum_{T_i>T} Z(T_i)$ are relatively small;

2b. Modify (a) so that the summing is over all $T_i<T$ and the first point, T^* say, exceeding or equal to T;

3. Sum over all T_i, $i \leqq M$, where the integer M is chosen so that the mean and variance of e_{T_M} are relatively small.

All these methods have drawbacks and advantages. For simplicity we consider in detail only method 2a. It has for example the drawback that the truncated sum X_T may become 0 (which is avoided by a use of 2b and X_{T^*}) but also the advantages:

(i) The tail e_T is independent of X_T and its distribution admits often a normal approximation (as we shall see);

(ii) The T_i's such that $T_i \leqq T$ can easily and rapidly be simulated by generating a Po (λT)-variable ν and by setting $T_i = TU_i$, $i=1,2,\ldots,\nu$, where the U_i's are random numbers from $U(0,1)$ (the order of the T_i's is irrelevant).

Under the assumption $H(y;u)=H(y/g(u))$, the LT of the tail e_T becomes

$$\varphi_{e_T}(s)=\exp\left\{\lambda\int_{(T,\infty)}(E[\exp\{-sg(u)V\}]-1)\,du\right\}, \tag{4.2}$$

where V stands for a random variable with d.f. H. If $E[V^2]<\infty$, we have, as $s\to 0$,

$$E[\exp\{-sV\}]=1-E[V]s+\frac{E[V^2]}{2}s^2+o(s^2).$$

Setting then $\mu_T=\lambda E[V]\cdot\int_{(T,\infty)} g(u)\,du$ and $\sigma_T^2=\lambda E[V^2]\cdot\int_{(T,\infty)}(g(u))^2\,du$, we easily find from (4.2) (or its characteristic function version) that e_T has

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asymptotically a normal distribution $N(\mu_T, \sigma_T^2)$ provided that

$$\sigma_T^{-1} \sup_{u>T} g(u) \to 0 \quad \text{as} \quad T \to \infty \tag{4.3}$$

(equivalently, $g(T)/\sigma_T \to 0$ if g is non-increasing). If we let H be degenerate at $y=1$ and set $\lambda = 1$ and $g(u) = \sup\{x; \bar{N}(x) > u\}$, then (4.3) reduces to the condition that $\int_{(0,\varepsilon)} y^2 N(dy)/\varepsilon^2 \to \infty$ as $\varepsilon \to 0$ which holds if N has a density $\boldsymbol{n}$ and $\varepsilon \boldsymbol{n}(\varepsilon) \to \infty$ as $\varepsilon \to 0$.

If $g(u) = e^{-u}$, then (4.3) is not satisfied, which is a little bit unfortunate. However, if the rate λ of the Poisson process is high, a normal approximation may still be good. In any case, we can always approximate e_T by its expected value μ_T. If $g(u) \sim Cu^{-\beta}$, $\beta > 1$, as $u \to \infty$, then (4.3) holds.

Concerning method 2b, e_{T^*} and X_{T^*} are conditionally independent given T^*. Given that $T^* = t$, the distribution of e_{T^*} coincides with that of e_t and a normal approximation may be possible. However, the fact that μ_{T^*} and $\sigma_{T^*}^2$ must be calculated for each T^* can give problems. This remark is relevant also for method 3 with T^* replaced by T_M.

If we let H be degenerate at 1, set $g(u) = \sup\{x; \bar{N}(x) > u\}$, and use X_T as our random number with the T_i's simulated by the technique in (ii) above ($\lambda = 1$), then this simulation method coincides with the method described at the beginning of Section 2 as is not hard to see.

The Poisson variable ν in (ii) ($\lambda = 1$) may be approximated by $\nu = T$ (here we assume that T is an integer). This corresponds to replacing X_T by a variable $\tilde{X}_T$ with LT

$$\varphi_{\tilde{X}_T}(s) = \left(T^{-1} \int_{(0,T)} \zeta(s; u)\, du\right)^T = \left(1 + T^{-1} \int_{(0,T)} (\zeta(s; u) - 1)\, du\right)^T$$

which of course tends to $\exp\{\int_{(0,\infty)} (\zeta(s; u) - 1)\, du\}$ as $T \to \infty$. Hence the approximation $\nu = T$ gives us an asymptotically correct random number $\tilde{X}_T$. In general the distribution of $\tilde{X}_T$ is no better an approximation of F than that of X_T. (We have $X = X_T + e_T$ and $\tilde{X}_T = X_T + e_T^*$, where e_T^* is independent of e_T and has mean 0.) This need not be the case for ID distributions on R.

4.2. We shall now as examples consider simulation from stable and gamma distributions.

Consider a stable distribution with LT $\varphi(s) = \exp\{-s^\alpha\}$, $0 < \alpha < 1$. The Lévy measure N has density $\boldsymbol{n}(y) = C_\alpha y^{-(1+\alpha)}$, where $C_\alpha = \alpha/\Gamma(1-\alpha)$. Hence $g(u) = \sup\{x; \bar{N}(x) > u\} = (C_\alpha/\alpha)^{1/\alpha} u^{-1/\alpha}$. So $X = (C_\alpha/\alpha)^{1/\alpha} \sum_{i=1}^\infty T_i^{-1/\alpha}$ is a random number from the given stable distribution if the T_i's are simulated from a Po(1)-process on $(0, \infty)$. If we truncate at T, we may replace the tail $e_T =$

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$X - X_T$ by a normal random number with mean

$$\mu_T = \int_{(T,\infty)} g(u)\, du = \frac{\alpha}{1-\alpha} (C_\alpha/\alpha)^{1/\alpha} T^{-(1-\alpha)/\alpha}$$

and variance

$$\sigma_T^2 = \int_{(T,\infty)} (g(u))^2\, du = \frac{\alpha}{2-\alpha} (C_\alpha/\alpha)^{2/\alpha} T^{-(2-\alpha)/\alpha}.$$

It remains to investigate how good this simulation method based on a normal approximation of the tail is. The shot-noise distribution is stable also for a non-degenerate shot-distribution but it is intuitively obvious that the efficiency cannot be improved by using such a shot-distribution.

If we instead consider a (for simplicity) symmetric stable distribution with characteristic function $\phi(t) = \exp\{-|t|^\alpha\}$, $0 < \alpha < 2$, the Lévy density is $\boldsymbol{n}(y) = C_\alpha |y|^{-(1+\alpha)}$, $y \in R \setminus \{0\}$, where $C_\alpha = (1/\pi)\Gamma(\alpha+1)\sin(\pi\alpha/2)$; cf., e.g., Exercise 8 in Chung (1974), pp. 158–159. (The corresponding constants in Feller (1971), pp. 569–570, are incorrect.) As response function we should take $g(u) = (C_\alpha/\alpha)^{1/\alpha} \operatorname{sign}(u)\, |u|^{-1/\alpha}$. The correct random number is $X = \lim_{T\to\infty} X_T$, where $X_T = \sum_{|T_i| \leq T} g(T_i)$ and the T_i's stem from a Po (1)-process on R. (The result that X is stable is due to Holtsmark; cf. Feller (1971), pp. 172–175.) The tail $e_T = X - X_T$ can approximately be replaced by a normal random number with mean $\mu_T = 0$ and variance

$$\sigma_T^2 = \int_{|u|>T} (g(u))^2\, du = \frac{2\alpha}{2-\alpha} (C_\alpha/\alpha)^{2/\alpha} T^{-(2-\alpha)/\alpha}.$$

If, for an integer T, we approximate X_T by

$$\tilde{X}_T = \sum_{i=1}^{2T} g(TU_i) = (C_\alpha/\alpha)^{1/\alpha} \sum_{i=1}^{2T} \operatorname{sign}(U_i)\, |U_i|^{-1/\alpha},$$

where the U_i's are simulated from $U(-1, 1)$, the simulation method reduces to the standard method of simulating stable random numbers by adding i.i.d. random numbers with appropriate tails of their distribution. This method is known to be good when the stable distribution is symmetric; cf. Bartels (1978). See Chambers et al. (1976) for a completely different elegant method of simulating stable random numbers.

The suggestion here that a normal approximation can be of help when one wants to simulate stable random numbers seems to be new. One does not expect that such an approximation could be useful since the stable distributions ($\alpha \neq 2$) have no normal factors.

We consider now simulation from a $\Gamma(p, 1)$-distribution (shape p, scale 1).

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The Lévy density equals pe^{-y}/y, $y>0$. So the random number

$$X=\sum_{i=1}^{\infty} g(T_i), \tag{4.4}$$

where $g(u)=\sup\{x; \int_{(x,\infty)} e^{-y}/y\, dy > u\}$ and the T_i's stem from a Po(p)-process on $(0,\infty)$, has a $\Gamma(p, 1)$-distribution. It is easy to see that $g(u)\sim e^{-u}$ as u tends to ∞. If $p\leqq 1$, which is the only case we really need consider, the convergence of the sum (4.4) is rapid but we cannot make a normal approximation of its tail. The disadvantage of this method is that it requires a computation of the inverse of the function $\int_{(x,\infty)} e^{-y}/y\, dy$. Note, however, that this inverse does not involve p and therefore the method is not that troublesome as the standard method that uses the inverse of the $\Gamma(p, 1)$-d.f.

To avoid the computation of the inverse, we may instead use exponential shot-distributions. We have that, with the T_i's as above,

$$X=\sum_{i=1}^{\infty} \exp\{-T_i\}V_i$$

is $\Gamma(p, 1)$-distributed if the V_i's are $\Gamma(1, 1)$ random numbers. The sum converges rapidly if $p\leqq 1$ and it should be possible to use the approximation X_T for a fairly small value of T (measured in multiples of $1/p$). We may add the mean pe^{-T} of e_T. To avoid X_T vanishing we may instead use X_{T^*} (see Section 4.1) and add the conditional mean $p\exp\{-T^*\}$ of e_{T^*} given T^*. In fact, if p is small, $\exp\{-T_1\}V_1$ is approximately $\Gamma(p, 1)$-distributed.

This method can hardly compete in speed with the (by now standard) method based on rejection technique, but it is extremely easy to program and therefore an interesting alternative.

4.3. We now turn to simulation from a g.g.c. with given Thorin measure U. (Many standard distributions, e.g. the stable and PF_∞ distributions, are g.g.c.'s but often it is hard to find explicitly the measure U.) We use the response function $g(u)=e^{-u}$ and appropriate shot-distributions. Since a variable with LT

$$\exp\left\{\int_{(T,\infty)} \log\left(\frac{1}{1+(s/\tau)}\right)U(d\tau)\right\}$$

can be approximated by a normal variable with mean $\mu_T=\int_{(T,\infty)} \tau^{-1}U(d\tau)$ and variance $\sigma_T^2=\int_{(T,\infty)} \tau^{-2}U(d\tau)$ if $T\sigma_T\to\infty$ as $T\to\infty$, we need most often only simulate from a g.g.c. with finite measure U. We then set $\lambda=U((0,\infty))$ and $M(d\tau)=\lambda^{-1}U(d\tau)$ and simulate points T_i from a Po(λ)-process on $(0,\infty)$ and i.i.d. random numbers τ_i from the (mixing) distribution M. The desired random number is then given by

$$X=\sum_{i=1}^{\infty} \exp\{-T_i\}V_i,$$

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where $V_i = \tau_i^{-1} W_i$ and the W_i's are simulated from $\Gamma(1, 1)$; cf. (3.5). Of course, the sum must be truncated.

As an application of the above, we give an approximate solution of a problem mentioned by McKenzie (1982). He found that for $0 \leqq \beta \leqq 1$ and $Y_0 \sim \Gamma(p, 1)$, there are i.i.d. random variables δ_n, $n = 1, 2, \ldots$, such that also Y_n, $n = 1, 2, \ldots$, defined recursively by $Y_n = Y_{n-1}^{\beta} \delta_n$, are $\Gamma(p, 1)$-distributed. As the distribution of δ_n is given only in terms of its Mellin transform, the problem arises how to simulate easily δ_n and the process.

We may set $X_n = -\log Y_n$, $\varepsilon_n = -\log \delta_n$ and the model transforms into an AR-1 model $X_n = \beta X_{n-1} + \varepsilon_n$, $n = 1, 2, \ldots$. It is known and not hard to show that X_n has the same distribution as

$$\sum_{k=0}^{\infty} \frac{Z_k - 1}{k+p} + C,$$

where $C = \lim_{m \to \infty} (\log m - \sum_{k=0}^{m} (k+p)^{-1})$ and $Z_1, Z_2, \ldots$ are i.i.d. $\Gamma(1, 1)$-variables. (The sum of random variables converges almost surely since $\sum_{k=0}^{\infty} (k+p)^{-2} < \infty$.) The distribution of X_n is an extended g.g.c. but we do not use that. The calculated value of C can be subtracted and the problem reduces to a simulation of an AR-1 model with marginals distributed as

$$\sum_{k=0}^{\infty} \frac{Z_k - 1}{k+p}.$$

It is easy to prove that, as $m \to \infty$,

$$\sum_{k=m}^{\infty} \frac{Z_k - 1}{k+p}$$

is asymptotically normal with mean 0 and variance

$$\sum_{k=m}^{\infty} (k+p)^{-2} \, (\sim (m+p)^{-1}).$$

The skewness is of order $(m+p)^{-\frac{1}{2}}$. Since an AR-1 model with normal marginals is easy to simulate, we see that an approximate solution is obtained if we can simulate an AR-1 model with marginals distributed as

$$\sum_{k=0}^{m-1} \frac{Z_k}{k+p}.$$

The distribution of the last variable is a g.g.c. with a measure U that has mass 1 at all the m points $p, p+1, \ldots, p+m-1$. If we now let $\{T_i\}_{-\infty}^{\infty}$ be points in a Po (m)-process on R and set

$$X(t) = \sum_{T_i \leqq t} \exp\{-(t - T_i)\} V_i,$$

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where $V_i = \tau_i^{-1} W_i$ with $W_i \sim \Gamma(1, 1)$ and the τ_i's uniformly distributed over the m points above, then $X(t)$ is distributed as

$$\sum_{k=0}^{m-1} \frac{Z_k}{k+p}$$

for every t. Now we choose ρ such that $\beta = e^{-\rho}$ and set $X_n = X(n\rho)$. Then we have

$$X_n = \beta X_{n-1} + \sum_{(n-1)\rho < T_i \leq n\rho} \exp\{-(n\rho - T_i)\} V_i, \qquad n = 1, 2, \ldots,$$

and we see that we only have to simulate X_0 (add exponential random numbers) and the T_i's (on $(0, \infty)$) and V_i's to get our approximate solution of the simulation problem. (An alternative method is to add simulated AR-1 processes with exponential marginals. Such processes are easy to simulate since the distributions of their innovations are simple mixtures of exponential distributions.)

5. Concluding comments

This paper is a product of my interest in infinite divisibility and simulation and my reading of the papers on AR processes by Lawrance and McKenzie mentioned in the text. The simulation method that has been described is often simple to program but not always rapid. Comparisons in speed have not been made. The Lévy measure is needed in a more or less explicit form.

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